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Assessing the effectiveness of the SOLIO Alfa Cure Plus device in treating low back pain: a randomized controlled study

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Abstract

Background Nonspecific low back pain (LBP) has become a significant worldwide public health problem. It is estimated that 84% of people present it at some point in their lives, in which 23% experience its chronic form, negatively affecting their daily lives. Because pain management tool that doesn't require a firm diagnosis, the development of a device, as SOLIO Alfa Cure Plus, that emanates low level laser therapy, radio frequency and heat with the goal of easing chronic back pain was highly expected.

Methods randomized, single blinded, controlled trial. Measures: Numeric Pain Rating Scale (NPRS), the Oswestry Disability Index (ODI) and the Schober's test. Thirty-seven patients completed pain, disability, and lower back flexibility scales. Randomization was obtained by having an equal amount of sham and real devices and distributing them randomly to patients out of a box where the devices were.

Results We observed a larger pain relief in the SOLIO group (42% vs. 23% $p=0.03$), and a higher improvement in flexibility (13%) compared to a worsening in the sham group (6.5%; $p=0.04$).

Conclusion We concluded that utilizing the SOLIO Alfa Cure Plus device may dramatically reduce back pain and allow patients to experience an improvement in quality of life as a result. *Trial registration*: The clinical trials registration is 8475-21-SMC.

Keywords Nonspecific low back pain, SOLIO Alfa Cure Plus device, Diagnosis, Pain visual analog scale, Radiofrequency, LLLT devices

Background

Low back pain (LBP) is a very frequent condition in the United States, affecting approximately 80% of adults. It is one of the most prevalent musculoskeletal complaints and a major contributor to disability worldwide [1]. More than 80% of people worldwide will suffer from LBP at some point in their lives, affecting between 4 and 33% of the population at any given moment. One-half of adults in the United States report having suffered LBP in the previous year, and one quarter report having suffered it in the past three months [2]. It is one of the leading causes of disability and missed workdays, with an estimated

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yearly cost of over $100 billion in healthcare expenses and lost productivity [3].

There is no universally accepted definition of sub-chronic LBP, but it is often considered as a pain lasting longer than three months and less than 6 months, while its acute form is considered to last less than three months [4]. Between 2% and 25% of people with acute LBP will later develop chronic LBP. Conservative treatments include pharmaceuticals, manual therapy (massage, physiotherapy, spinal manipulation), exercise therapy [5] (aerobic activity, muscular strengthening), and educational or psychological therapy (cognitive behavioral therapy, support groups, educational sessions) [6].

Clinical practice recommendations for the diagnosis and treatment of acute and chronic LBP were developed by the American Pain Society and the American College of Physicians. The use of non-steroidal anti-inflammatory drugs, muscle relaxants, and the application of superficial heat were the only three treatments with excellent evidence, suggesting at least a modest effect on acute LBP [7]. Exercise therapy, physiotherapy, cognitive behavioral therapy, interdisciplinary rehabilitation, spinal manipulation, and tricyclic antidepressants presented excellent effectiveness evidence for persistent LBP [8, 37, 38]. Furthermore, there are modern studies aimed to provide evidence for or against common treatments for LBP. Recent studies have been made on acupuncture [29], mesenchymal stem cells [30], gabapentinoids [32] and prolotherapy [34] are just some of the modalities being more thorughly studied today, those might indicate the need for better treatment to become available.

Studies have shown that some generally recommended first-line medications for LBP are only slightly more effective than placebo controls in double-blind randomized trials [9], such as the epidural corticosteroid injection that also presents a complication rate between 0% and 9.65%. Another modality is pharmacologic treatment, a wide range of medicines is available to help manage LBP while none are universal and should be accompanied by other modalities, highlighting the need for individualized treatment [31, 33, 36, 38]. Another treatment option is physical exercise in combination with manual therapy that often produces short-term results [10, 39]. Various energy sources are thought to have great potential for both general healing and pain relief [11] and there are several available devices that make use of this technology [12].

Infrared therapy (IR) can reach muscles, nerves and even bones, where it reduces inflammation by accessing inner layers of tissues. The use of IR therapy has been shown to accelerate wound healing, ease arthritis knee pain, increase endorphin levels, and activate neuromodulators without any side effects [13]. According to a recent study, IR therapy is beneficial for chronic non-specific LBP sufferers in terms of comfort, function, lumbar range of motion, and back extensor endurance [14].

Radio frequency (RF) therapy shows promise as a healing strategy in a variety of etiologies [15–17] and it one of the most common energy sources for producing therapeutic heat levels. RF energy is delivered to soft tissue as part of the mechanism's biophysical therapeutic mode to relieve pain, as well as enhancing collagen formation, vascularization, and collagen structure in the tissue [15–17]. Another energy source is the tow-level laser treatment (LLLT), known to induce analgesia, reduce inflammation, and promote healing in a variety of musculoskeletal diseases. This technique is based on the theory that light causes biochemical changes in cells like photosynthesis in plants [18]. LLLT may also be used as an adjuvant therapy for pain management in patients with neurologic impairments and neuropathic pain [19].

The techniques are potential tools for controlling pain in a variety of etiologies, but additional research appears to be needed to fully develop the therapeutic and mechanistic role of these techniques. No device combining all three approaches has ever been objectively assessed in a randomized clinical trial for pain management. The SOLIO Alfa Cure Plus is the only home-use device with FDA clearance that uses the proprietary synergy of three energy sources (IR, RF, and LLLT), and this study aims to assess the SOLIO Alfa Cure Plus's efficiency in providing pain relief. This technology is the first noninvasive and patented home RF device. SOLIO Alfa Cure Plus uses continuous RF, the internal controller reduces the RF power while reaching the treatment temperature of 39–42 Celsius.

These three treatment modalities are hypothesized to affect the physiological issue at many levels, causing tissue regeneration and pain relief at the cellular and tissue levels, as well as changing sensory nerves and cell molecular makeup [14, 19]. Our goal was to determine whether the RF component of the treatment has an effect on back pain relief.

Patients/methods

Study population

Forty patients with sub-chronic LBP, lasting at least 6 weeks and less than 6 months (score 4 or higher according to the NRS) between 18 and 70 years old, that possessed assistance of family members, as the device is to be used on the back which is hard to accomplish without help. This study excluded patients with fibromyalgia, pregnancy, pacemaker implant, hyper coagulation, cognitive impairment, psychosis diagnosis, addiction, opiate usage, malignant cancer, and BMI > 35 or 18.5. The participants were patients of the Institute for Pain Medicine at Chaim Sheba Medical Center, who had a previous diagnosis of sub-chronic LBP.

Study design

This is a randomized, prospective, single-center comparative sham-controlled trial examining the effectiveness of the SOLIO Alfa Cure Plus Device for patients with LBP. The clinical trials registration is 8475-21-SMC.

Instruments

The Pain Numerical Rating Scale (NRS) - it is an 11-point scale for self-report of pain. It is the most used unidimensional pain scale. The respondent selects a whole number (integers 0–10) that best reflects the intensity (or other quality if requested of his/her pain.

The OWESTRY questionnaire [20] - was completed once per week to evaluate the current condition of the participant's back pain. The ODQ is comprised of 10 questions about the participant's daily experience with their back pain and how it affects diverse day to day activities (e.g., social life, work life, walking and standing, etc.). Every question has 6 different answers (given scores from 0 to 5) to choose from ranging from according to the effect of the back-pain on that specific activity and a disability score is calculated from those answers and recorded. A disability score of < 5 suggests no disability, 5–14 suggests "mild disability", 15–24 is "moderate disability", 24–34 is "Severe disability" and finally 35–50 suggests "Complete disability".

The Schober's test [21] - The test involves measuring the distance between two points on the lower back: one at the level of the posterior superior iliac spines (PSIS) and the other 10 centimeters above the PSIS. The measurements are taken both while the participants were standing upright and during maximum forward bending. The researcher uses a tape held over the spine by marking a point 5 cm below and 10 cm above the lumbosacral junction. A measurement of > 5 is considered normal while a measurement < 5 may indicate a decrease in flexibility. It must be completed at the beginning and at the end of the study. To increase the reliability of the Schober's test the research team ensured that the same person made all the measurements for all the patients using the same measuring tape [27, 28].

The SOLIO Alfa Cure Plus Device for patients with LBP- is a round device equipped with a strap for hand holding, it is to be used while connected to a socket. The device is used by an assistant (usually a family member) on the participant's back for 20 min while moving it around in circles on the lower back. It should be mentioned this is the first trial in Home-RF using an external. All the devices' specifications are depicted in Table 1.

Table 1 Demographics

group	Total	Male	Female	Age - Mean (SD)
real	18	8	10	47.8 (15.5)
sham	17	12	5	45.6 (14.04)

Procedure

We recruited patients on their first arrival to the institute for Pain Medicine who suffered from LBP for 3 or more months at the time, each patient was given a thorough explanation on the device's kit and watched an instructional video.

A total of 40 devices were randomly (simple randomization – recruiters/researchers assigned devices out of a box where devices were shuffled) assigned to patients participating in the trial, 20 of which were active and 20 of which were sham devices. After enrollment, one patient was excluded due to an epidural steroid injection, and two other patients were excluded due to noncompliance with the trial's treatment protocol. There were 20 men and 17 women among the study patients (54.1% men and 45.9% women).

To ensure all patients received the same instructions and to increase compliance, a video with clear instructions on how to use the device was provided. Using the device independently for LBP is not possible, so the patient needs an "assistant" to apply the device for 20 min on their back.

Statistical analysis

The results of the measures were analyzed using SPSS software. We conducted three distinct clinical evaluations based on patient data gathered about SOLIO Alfa Cure Plus devices. Using Student's T-test (paired), responses were assessed and compared between the active group and the sham group to get *p*-values and establish whether there was any difference in the responses from each group.

Results

The study end point included 37 patients (males and females from age 18–70 years) divided into two groups: 18 (average age – 55 years) were active and 19 (average age – 48 years) used sham devices.

Table 2 depicts the average results from all three clinical evaluations. While the initial NRS average for the sham-group was 5.58 (± 2.15), and the initial NRS average for the active-group was 6.29 (± 1.79), after 4 weeks of treatment, both the active and sham groups' average decreased to 4.29 (± 2.05) and 3.64 (± 1.80) respectively. The mean result of the Schober test for the sham-group was 5.04 (± 1.55) and the mean for the active-group was 4.57 (± 1.70). There was a decrease in the Schober's test results to 4.76 (± 1.25) in contrast to the active group, where an increase in the Schober test score to 5.18 (± 1.61) was found.

As seen in Table 3, The mean of the ODQ score was for the sham-group was 15.49 (± 7.42) and for the active-group was 17.05 (± 8.47) After 4 weeks of treatment, the sham-group ODQ decreased to 14.88 (± 8.44) and the

Table 2 Device specifications

LLLT	
Total radiation	60–95 mW
IR Radiation	90 mW
RED Radiation	> 50 mW
IR wavelength	805–915 nm
RED wavelength	620 nm
Far IR	
Heating	10 W
RF	
Power	24 W
Frequency	1 MHz
Additional information	Focus 10 mm wide will be 2–3 cm and can spread the heat deeper into skin layers by consistently focusing the heat on a certain area
Heating temperature	39–42 °C

*IR/RED = InfraRed

active-group decreased to 14.35 (± 8.11), There was no statistically significant difference in ODQ scores, but when we compared the 2 groups' sleep criteria in ODQ individually, we found a statistically significant difference ($p = 0.034$).

Discussion

Currently, we seek ways to mitigate the suffering caused by generalized back pain, which is a prevalent phenomenon in the modern age. According to the study's findings, patients who frequently used the SOLIO Alfa Cure Plus device had significant less pain than those who frequently used RF-uncapable devices. Helmhout et al. (2008) [22] conducted a randomized controlled experiment in military recruits to assess the effectiveness of device-supported specialized lumbar extensor training compared with routine physical therapy. They reported that for the treatment of LBP, device-supported targeted back strengthening was not significantly more effective than standard physiotherapy.

The device-supported therapy employed in Helmhout et al. (2008) [22] differs from the protocol used in our study, which was modified according to variability in strength, motion, and function as indicated by the David Spine Concept system. Strengthening of lumbar extensors was the primary focus of their procedure [23]. Based on studies that are related to our own, proper treatment for LBP should include not only pain reduction but also nerve function restoration [23, 24].

Additionally, while comparing the two group's sleep criteria in the ODQ, we found a significant result ($p = 0.034$) after sub-analyzing each of the ODQ's items separately. Considering how good sleep quality frequently enhances one's quality of life and ability to manage pain, we should investigate whether the SOLIO Alfa Cure Plus device may contribute to patients' increased satisfaction with their sleep habits. Schober's test results indicate that the active group had a greater degree of flexibility after using the device, which may indicate that RF technology is effective in improving flexibility. The interpretation of the research findings should be done with caution due to its limitations. First, because most of the trial was conducted at home, it was difficult to track patients' compliance and how they used the SOLIO Alfa Cure Plus device. In order to overcome this difficulty, we periodically checked in with participants and asked them about the device use. Another limitation was the inability to account for regression to the mean in a study with diverse population, (e.g., everyone suffers from back pain but have different personal "profiles") this statistical effect could affect our study as the Sham-group also improved on the NRS and ODQ criteria as expected even without any intervention.

This data will prove useful to an inexperienced clinician who has recently been trained in using the SOLIO Alfa Cure Plus device for LBP and wish to improve their learning based on the practice patterns of other clinicians. Furthermore, this study may provide insight into the general practice trends in the field of pain medicine and motivate clinicians to make changes to their current practice. As a result of this, future research efforts to examine the relationship between positive analgesic outcomes and variables related to providers, such as the number of diagnostic blocks, the type of SOLIO Alfa Cure Plus device, the amount of contrast utilized

Table 3 Group results comparison

Groups comparison							
		Sham ($n = 19$)		**Active ($n = 18$)**			
		t = 0	**t = 4 weeks**	**t = 0**	**t = 4 weeks**	**p1**	**p2**
NRS	Mean (range)	5.58 (± 2.15)	4.29 (± 2.05)	6.29 (± 1.79)	3.64 (± 1.80)	0.3681	0.0316
Schober test (cm)	Mean (± SD)	5.04 (± 1.55)	4.76 (± 1.25)	4.57 (± 1.70)	5.18 (± 1.61)	0.5823	0.0477
ODQ score	Median (range)	15.94 (± 7.42)	14.88 (± 8.44)	17.05 (± 8.47)	14.35 (± 8.11)	0.3789	0.7414

Abbreviations: NRS, Numeric Rating Scale; ODQ, Owestry Disability Questionnaire

P1 - comparison of sham and active groups at t = 0

P2 - comparison of sham and active groups at t = 4 weeks

during diagnostic blocks, and many others [25] might be performed.

To treat sub-chronic, nonspecific LBP, numerous invasive procedures must be performed. These typically begin after the problem persists for a period of time [26]. Radio-frequency lesioning and injections into the facet joints are two procedures that are commonly performed along pharmacological injections in pain clinics [35]. It has been shown that discomfort in the facet joints contributes to LBP, although the function of therapeutic approaches is still unclear [26]. Our findings suggest that the SOLIO Alfa Cure Plus device may significantly reduce back pain and improve quality of life for patients. Therefore, we recommend investing more resources in noninvasive RF pain management. Clinical evidence suggests that SOLIO Alfa Cure Plus has a strong therapeutic effect on chronic non-specific low back pain when considered collectively.

Conclusion

There are many people who experience generalized back pain, and it is important to provide them with support so they can maintain their quality of life despite the discomfort. Our findings suggest that the SOLIO Alfa Cure Plus device can significantly reduce back pain and improve quality of life for sub-chronic low back pain patients; as a result, we recommend allocating more resources to the study of non-invasive RF pain relief.

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Author contributions
Conceptualization- L.U. Data curation- Y.H. Formal analysis- Y.H. Funding acquisition-L.U. Project administration- G.P. Resources-G. PSupervision- I.G. Writing - original draft-Y.H. Writing - review & editing - L.U.

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Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
All patients gave informed consent to participate in the study. The Sheba Medical Center Ethics Committee approved the study. The clinical trials registration is 8475-21-SMC.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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